

◆ AWS Solutions Architect Associate (SAA-C03)

◆ Learning Notes

Instance Types, High Availability, Fault Tolerance & Scaling Concepts

◆ 1. Understanding EC2 Instance Types

What is an Instance Type?

An EC2 instance type defines the amount of:

- CPU (Compute Power)
- Memory (RAM)
- Storage
- Network Performance

available to a virtual server.

Think of an instance type as selecting the specifications of a computer before purchasing it.

Why AWS Has Different Instance Types

Different workloads require different resources.

Examples:

- Web servers need balanced resources
- Databases need large memory
- Analytics workloads need high CPU power
- Big data workloads need fast storage

AWS provides specialized instance families to optimize cost and performance.

◆ 2. General Purpose Instances

Purpose

Balanced CPU, Memory, and Networking.

Common Families

- T3
- T4g
- M5
- M6i

Best For

- Web applications
- Development environments
- Small databases
- Business applications

Example

A company hosts a WordPress website.

Recommended Instance:

`t3.micro`

because it provides balanced resources at a low cost.

Exam Tip

If the question states:

Balanced compute, memory, and networking

Choose:

General Purpose Instance

◆ 3. Compute Optimized Instances

Purpose

High CPU performance.

Common Families

- C5

- C6i
- C7g

Best For

- Machine Learning
- Scientific Modelling
- Gaming Servers
- Batch Processing

Example

A company performs large mathematical calculations.

Recommended Instance:

`c5.large`

Exam Tip

Keywords:

- Compute-intensive
- High processing
- CPU-heavy

Answer:

`Compute Optimized`

◆ 4. Memory Optimized Instances

Purpose

Large RAM capacity.

Common Families

- R5
- R6i
- X1

Best For

- SAP HANA
- In-memory databases
- Redis

- Enterprise databases

Example

A company runs a 2TB database entirely in memory.

Recommended Instance:

`r5.large`

or larger.

Exam Tip

Keywords:

- Database
- Memory intensive
- In-memory processing

Answer:

`Memory Optimized`

◆ 5. Storage Optimized Instances

Purpose

Fast local storage access.

Common Families

- I3
- D2
- H1

Best For

- Big Data
- Data Warehouses
- High-speed analytics

Example

A company processes terabytes of log files daily.

Recommended Instance:

i3.large

Exam Tip

Keywords:

- Massive storage
- High IOPS
- Data warehouse

Answer:

Storage Optimized

◆ 6. High Availability (HA)

Definition

High Availability means a system remains accessible and operational even when failures occur.

Goal:

Minimize downtime

AWS High Availability Design

A highly available application is deployed across multiple Availability Zones.

Example:

AZ-A

AZ-B

AZ-C

If one zone fails, traffic is redirected to healthy zones.

AWS Services Supporting HA

Elastic Load Balancer (ELB)

Distributes traffic across multiple servers.

Auto Scaling

Adds or removes servers automatically.

Route 53

Provides DNS failover capabilities.

Multi-AZ RDS

Creates standby database replicas.

Exam Tip

When the exam mentions:

Improve availability

Look for:

- Multiple Availability Zones
 - Load Balancers
 - Auto Scaling
-

◆ 7. Fault Tolerance

Definition

Fault Tolerance means the application continues operating even when components fail.

Fault Tolerance is stronger than High Availability.

High Availability vs Fault Tolerance

High Availability

Some downtime may occur.

Example:

99.99% uptime

Fault Tolerance

No interruption should occur.

Example:

Automatic failover with no user impact

AWS Fault-Tolerant Design

Load Balancer



EC2 Instance A

EC2 Instance B

EC2 Instance C

If Instance A fails:

Traffic automatically moves to B and C.

Users remain unaffected.

Exam Tip

Keywords:

- Continue operating
- Survive failure
- No interruption

Answer:

Fault Tolerant Architecture

◆ 8. Vertical Scaling

Definition

Increasing resources of a single server.

Also called:

Scale Up

Example

Before:

`t3.micro`

After:

`t3.large`

Resources increase:

- More CPU
- More Memory
- Better Performance

Advantages

✓ Easy

✓ Quick

✓ Simple implementation

Disadvantages

✗ Hardware limits exist

✗ Downtime may be required

✗ More expensive at scale

◆ 9. Horizontal Scaling

Definition

Adding additional servers.

Also called:

Scale Out

Example

Before:

2 EC2 Instances

After:

10 EC2 Instances

Traffic is distributed using a Load Balancer.

Advantages

- ✓ Highly available
 - ✓ Fault tolerant
 - ✓ Elastic
 - ✓ Preferred AWS architecture
-

Disadvantages

- ✗ More complex architecture
 - ✗ Requires load balancing
-

Exam Tip

AWS generally prefers:

Horizontal Scaling

over

Vertical Scaling

◆ 10. Auto Scaling Groups (ASG)

Definition

Auto Scaling automatically adjusts the number of EC2 instances.

Goal:

Match capacity with demand

How Auto Scaling Works

Traffic Increases

↓
CPU > 70%

↓
Launch New EC2

Traffic Decreases

↓
CPU < 20%

↓
Terminate EC2

Benefits

High Availability

Applications remain online.

Cost Optimization

Pay only for resources needed.

Elasticity

Scale automatically.

Fault Tolerance

Failed instances are replaced automatically.

◆ SAA Exam Scenario

Question

An e-commerce website experiences traffic spikes during Black Friday.

Requirements:

- High Availability
- Fault Tolerance
- Automatic Scaling
- Cost Optimization

Best Solution

Application Load Balancer
+
Auto Scaling Group
+
Multiple Availability Zones

Why?

Load Balancer

Distributes traffic.

Auto Scaling

Adds capacity automatically.

Multiple AZs

Provides High Availability and Fault Tolerance.

◆ Key Exam Takeaways

General Purpose

Balanced workloads.

Compute Optimized

CPU-intensive workloads.

Memory Optimized

Database and memory-intensive workloads.

Storage Optimized

High-speed storage workloads.

High Availability

Minimize downtime.

Fault Tolerance

Continue operating after failure.

Vertical Scaling

Scale Up.

Horizontal Scaling

Scale Out.

Auto Scaling

Automatic capacity management.

◆ Quick Memory Trick

Instance Types

G-C-M-S

- **G** → General Purpose
- **C** → Compute Optimized
- **M** → Memory Optimized
- **S** → Storage Optimized

Scaling

Up = Bigger Server

Out = More Servers

Availability

HA = Stay Online

Fault Tolerance

FT = Survive Failure

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